

CHAPTER 5

OSI and TCP/IP Models

Layered architecture · seven OSI layers · encapsulation · TCP/IP four layers · TCP vs UDP · ports and addresses · OSI vs TCP/IP

QUICK TRICK

Learn the fast tricks first (details below):

1. **OSI bottom-up:** Please **Do Not Throw Sausage Pizza Away** = Physical, Data link, Network, Transport, Session, Presentation, Application.
2. **Data units bottom-up:** Bits → Frame → Packet → Segment → Data.
3. **Addresses:** Layer 2 = MAC · Layer 3 = IP · Layer 4 = Port · Layer 7 = URL / e-mail address.
4. **TCP/IP has 4 layers:** Application (OSI 5-7), Transport (4), Internet (3), Network Access (1-2).

5.1 Why do we use layers?

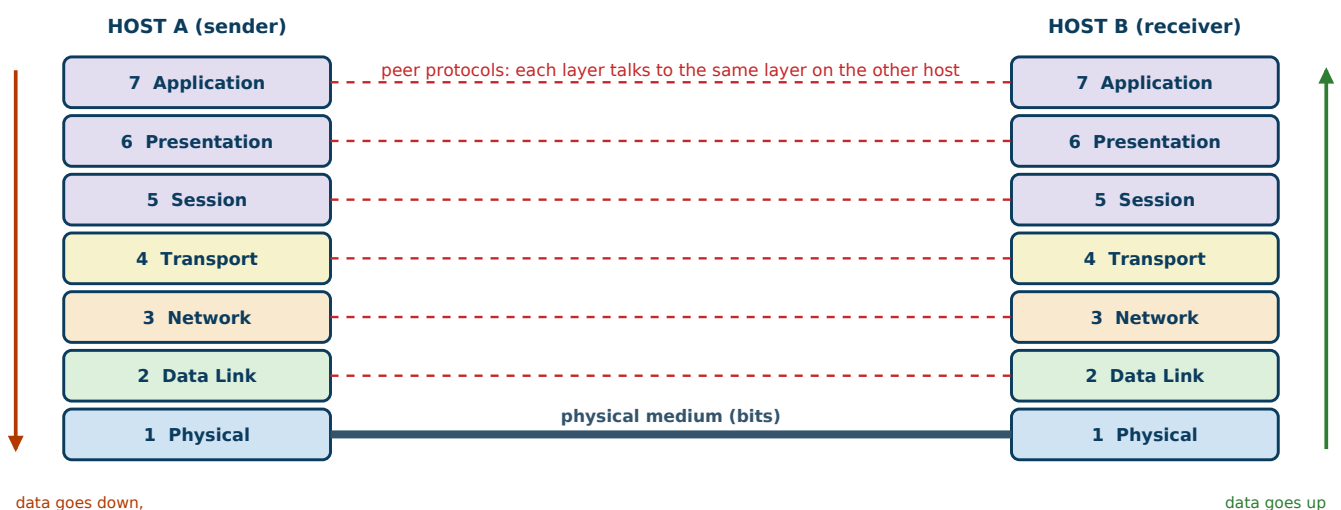
● LEVEL 1 · BEGINNER

The problem. Sending a file from your phone to a friend's computer across the world involves hundreds of tasks: turning the file into signals, finding the path, fixing errors, checking that the friend's computer is ready, and showing the file. One giant program doing all this would be very hard to build and change. **The idea.** Divide the job into **layers**. Each layer does **one group of tasks**, uses the services of the layer **below** and gives services to the layer **above**.

Analogy. Sending a letter. You write it (application), put it in an envelope with an address (network), the post office sorts and carries it (data link), and trucks and planes move it (physical). Your friend does the reverse steps. The postman does not need to read your letter, and you do not need to know how planes fly.

Advantage of layering	Meaning
Simple design	Each layer is a smaller problem
Modularity	A layer can be changed (for example Wi-Fi instead of cable) without touching the others
Standardisation	Companies build products for one layer that work with products of other companies
Easy troubleshooting	Find which layer has the fault ("Is it the cable or the website?")
Teaching	Everyone uses the same names and picture of a network

Key words: **Protocol** = rules between the **same layer** on two different computers (peer-to-peer). **Interface** = how a layer talks to the layer directly above or below it on the **same** computer. **Service** = what a layer offers to the layer above.



5.2 The OSI reference model

● LEVEL 1 · BEGINNER

OSI = Open Systems Interconnection. It was published by **ISO** (International Organization for Standardization) in **1984** as a **reference model**: a common plan, not a working product. It has **7 layers**.

Group	Layers	Role
User (application) support layers	7, 6, 5	Deal with the user's data and its meaning; mostly software
Transport layer (the link between groups)	4	Makes sure the whole message arrives end to end
Network support layers	3, 2, 1	Deal with moving data across the network; hardware and drivers

QUICK TRICK

Top-down mnemonic: All People Seem To Need Data Processing (Application, Presentation, Session, Transport, Network, Data link, Physical).

Bottom-up mnemonic: Please Do Not Throw Sausage Pizza Away.

Layer 1: Physical layer

- Moves **raw bits** as electrical, light or radio signals over the medium (Chapter 2).
- Defines cables and connectors, voltage levels, **data rate (bits per second)**, bit synchronisation, line configuration (point-to-point or multipoint), physical topology and transmission mode (simplex, half, full duplex).
- **Devices:** repeater, hub, modem, cables. **Data unit:** bits.

Layer 2: Data Link layer

- Makes the raw link reliable for the neighbouring node: **framing** (bits into frames), **physical (MAC) addressing**, **error control** (CRC, Chapter 4), **flow control**, and **access control** (who may use a shared medium: CSMA/CD in Ethernet, CSMA/CA in Wi-Fi).
- Two sublayers: **LLC** (Logical Link Control, IEEE 802.2) and **MAC** (Media Access Control).
- **Devices:** bridge, switch, NIC, access point. **Data unit:** frame. **Protocols:** Ethernet, PPP, HDLC, Frame Relay.

Layer 3: Network layer

- Moves **packets from source host to destination host across many networks: logical (IP) addressing, routing** (choosing the best path), packet forwarding, fragmentation, and congestion control.
- **Devices:** router, Layer-3 switch. **Data unit:** packet (datagram). **Protocols:** IP, ICMP, IGMP, routing protocols (RIP, OSPF, BGP), IPsec.

Layer 4: Transport layer

- **End-to-end, process-to-process** delivery between the right **programs**, using **port numbers** (the network layer only reaches the right computer).
- **Segmentation and reassembly** of messages, **connection control** (connection-oriented or connectionless), end-to-end **flow control** and **error control**.
- **Protocols:** **TCP** (reliable), **UDP** (unreliable but fast). **Data unit:** segment (TCP) or datagram (UDP).

Layer 5: Session layer

- Opens, manages and closes a **session** (a conversation) between two applications.
- **Dialog control** (who talks when: half or full duplex) and **synchronisation** by inserting **checkpoints**; if a transfer of 1000 pages fails at page 540, it resumes from the last checkpoint.
- Examples: NetBIOS, RPC, PPTP. **Data unit:** data.

Layer 6: Presentation layer

- Takes care of the **syntax and meaning** of data: **translation** between different codes (for example ASCII and EBCDIC, or different byte orders), **encryption / decryption** and **compression / decompression**.
- Formats: ASCII, Unicode, JPEG, GIF, PNG, MPEG, MP3. Encryption (SSL / TLS) is often placed here (or between 5 and 6). **Data unit:** data.

Layer 7: Application layer

- The layer closest to the user. It gives **network services to user programs**: file transfer, e-mail, web access, remote login, name lookup.
- Protocols**: HTTP, HTTPS, FTP, SMTP, POP3, IMAP, TELNET, SSH, DNS, DHCP, SNMP.
- The browser or e-mail program itself is **not** the application layer; it **uses** an application-layer protocol.

All seven layers on one page

#	Layer	Main functions (key words)	Data unit	Address / ID	Devices / examples
7	Application	User services: web, mail, file transfer	Data	URL, e-mail address	HTTP, FTP, SMTP, DNS; gateway
6	Presentation	Translation, encryption, compression	Data	—	JPEG, MP3, ASCII, SSL/TLS
5	Session	Dialog control, synchronisation, checkpoints	Data	—	NetBIOS, RPC
4	Transport	Process-to-process, segmentation, flow and error control	Segment	Port number	TCP, UDP
3	Network	Logical addressing, routing	Packet	IP address	IP, ICMP; router
2	Data Link	Framing, MAC address, error and flow control, media access	Frame	MAC address	Ethernet, PPP; switch, bridge, NIC
1	Physical	Bits, signals, media, data rate	Bits	—	Cable, hub, repeater, modem

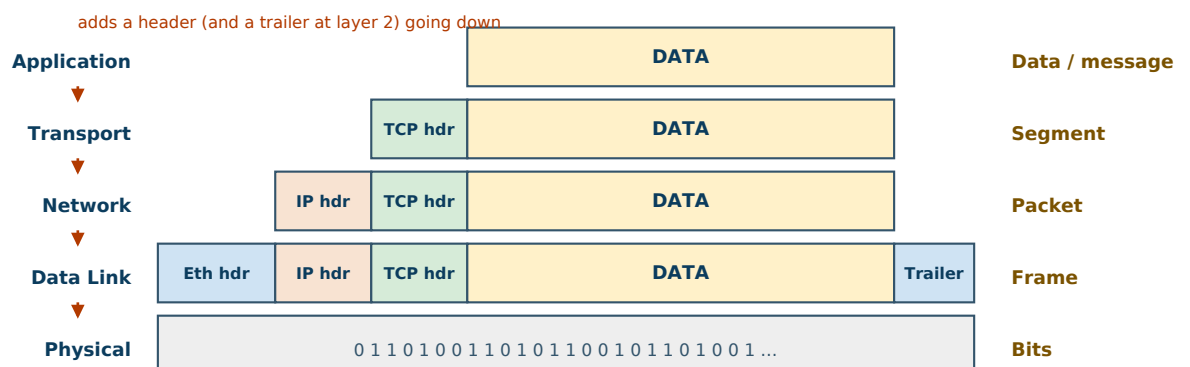
Which layer? Quick match

If the question says ...	Answer
Voltage, cable, bit rate, connector	Physical
Framing, MAC address, CRC / parity, CSMA	Data Link
IP address, routing, packet forwarding, congestion	Network
Port number, TCP / UDP, end-to-end, segmentation	Transport
Dialog control, checkpoint, session	Session
Encryption, compression, translation, JPEG	Presentation
HTTP, FTP, SMTP, DNS, e-mail, web page	Application
Flow control (hop by hop) and (end to end)	Data Link and Transport
Error control	Data Link and Transport

5.3 Encapsulation and decapsulation

● LEVEL 2 · EXAM

When data goes **down** the layers at the sender, each layer **adds its own header** in front of the data it receives (the layer-2 also adds a **trailer** with the CRC). This is **encapsulation**. At the receiver each layer **removes** its header and passes the rest **up**: **decapsulation**.



Term	Meaning
PDU	Protocol Data Unit: the name of the data at a layer (segment, packet, frame ...)
Header	Control information added in front (addresses, sequence numbers, flags)
Payload	The data a layer receives from the layer above, carried inside its PDU
MTU	Maximum Transmission Unit: biggest payload a link can carry in one frame (Ethernet = 1500 bytes)
MSS	Maximum Segment Size: biggest TCP data in one segment = MTU – IP header – TCP header

Example. Ethernet MTU = 1500 bytes, IP header = 20 bytes, TCP header = 20 bytes. **MSS** = 1500 – 20 – 20 = **1460 bytes**. An Ethernet frame adds a 14-byte header and 4-byte trailer (18 bytes), so the largest frame = 1460 + 20 + 20 + 18 = **1518 bytes**.

Example (overhead). Send 3000 bytes of data with TCP/IP over Ethernet. Number of segments = $\lceil 3000 \div 1460 \rceil = 3$ (1460 + 1460 + 80 bytes). Each adds 20 + 20 + 18 = 58 bytes of headers and trailer, so overhead = $3 \times 58 = 174$ bytes and total bytes sent = 3000 + 174 = **3174** (preamble ignored).

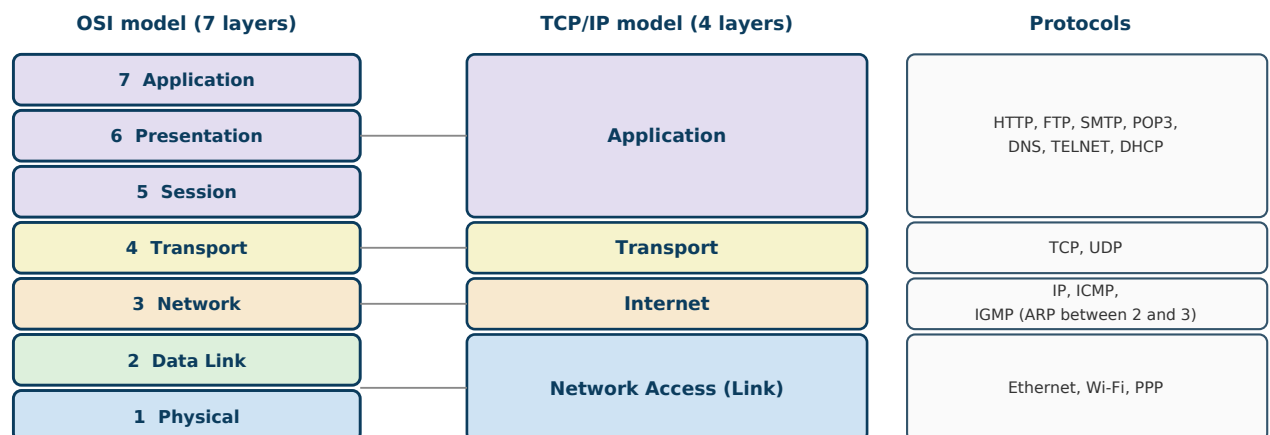
QUICK TRICK

Down = **add** headers (encapsulation). Up = **remove** headers (decapsulation). Only layer 2 adds a **trailer**.

5.4 The TCP/IP model

● LEVEL 1 • BEGINNER

TCP/IP was developed by **DARPA** for the ARPANET (Chapter 1) and is the model the Internet actually uses. The protocols were built first, and the model was written afterwards to describe them. It has **4 layers** (some books show 5 layers by splitting the bottom layer into Data Link and Physical).



TCP/IP layer	OSI layers covered	Job	Main protocols
Application	7, 6, 5	User services; also handles data format and sessions	HTTP, HTTPS, FTP, SMTP, POP3, IMAP, TELNET, SSH, DNS, DHCP, SNMP, TFTP
Transport	4	End-to-end delivery between programs	TCP, UDP
Internet	3	Logical addressing and routing of packets	IP (IPv4, IPv6), ICMP, IGMP, routing protocols
Network Access (Link / Host-to-network)	2, 1	Sending frames over the local link	Ethernet, Wi-Fi, PPP, Frame Relay

BPSC POINT

ARP (Address Resolution Protocol) finds the MAC address that belongs to an IP address. It sits between layers 2 and 3, so books place it at the Internet layer or the link layer. **ICMP** (ping, error messages) works with IP at the Internet layer.

TCP/IP **Internet layer** = **OSI Network layer**; the word "Internet" here means the layer that joins many networks.

TCP and UDP

Point	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (handshake first)	Connectionless (just send)
Reliability	Reliable: ACKs, retransmission, no loss, no duplicates	Unreliable: no ACK, loss possible
Order	Data arrives in order (sequence numbers)	May arrive out of order
Speed	Slower (more work)	Faster (less work)
Flow and congestion control	Yes	No
Header size	20 bytes minimum	8 bytes (fixed)
Data unit	Segment	Datagram
Used by	Web (HTTP/HTTPS), e-mail, FTP, TELNET, SSH	DNS queries, DHCP, TFTP, VoIP, video streaming, online games, SNMP

TCP three-way handshake (opening a connection): **1.** Client sends **SYN**. **2.** Server replies **SYN + ACK**. **3.** Client sends **ACK**. To close, each side sends **FIN** and the other acknowledges it.

Example (sequence numbers). Client starts at 100, server at 300. Step 1: SYN, seq = 100. Step 2: SYN-ACK, seq = 300, ack = 101. Step 3: ACK, seq = 101, ack = 301. (The ack number is the next byte the sender expects.)

TCP header field	Size	UDP header field	Size
Source port	16 bits	Source port	16 bits
Destination port	16 bits	Destination port	16 bits
Sequence number	32 bits	Length (header + data)	16 bits
Acknowledgement number	32 bits	Checksum	16 bits
Header length, flags (URG, ACK, PSH, RST, SYN, FIN), window	4 + 6 + 16 bits (plus reserved)		
Checksum, urgent pointer	16 + 16 bits		

Port numbers (the address of a program)

A **port number** is a **16-bit** number (0 to 65535, so **65 536** ports). The IP address finds the computer; the port finds the **program** on it. **0-1023** are **well-known** ports, **1024-49151** registered, and **49152-65535** dynamic (temporary) ports.

Service	Port	Transport	Service	Port	Transport
FTP (data / control)	20 / 21	TCP	HTTP	80	TCP
SSH	22	TCP	POP3	110	TCP
TELNET	23	TCP	IMAP	143	TCP
SMTP	25	TCP	SNMP	161	UDP
DNS	53	UDP (and TCP)	HTTPS	443	TCP
DHCP (server / client)	67 / 68	UDP	RDP	3389	TCP
TFTP	69	UDP	NTP	123	UDP

QUICK TRICK

Remember the big four: **HTTP 80 · HTTPS 443 · FTP 21 · SMTP 25**. Then **DNS 53 · TELNET 23 · SSH 22 · POP3 110 · IMAP 143**. These protocols are explained in the next chapters.

5.5 Four levels of addresses

● LEVEL 2 · EXAM

Level	Layer	Address	Size	Identifies	Example
Physical	Data Link (2)	MAC address	48 bits	The network card on this link	00:1A:2B:3C:4D:5E
Logical	Network (3)	IP address	32 bits (IPv4)	A host on the network	192.168.1.10
Port	Transport (4)	Port number	16 bits	A program / service	80
Application	Application (7)	URL, e-mail address	—	A resource or user	www.example.com

BPSC POINT

As a packet crosses many routers, the IP addresses (source and destination) stay the same, but the MAC addresses change at every hop (source MAC = the device that just sent it, destination MAC = the next device). The port numbers also stay the same.

(Exception: NAT, a router that rewrites private IP addresses.)

Putting it together: opening a web page

Step	Layer	What happens
1	Application	Browser makes an HTTP request for the page. DNS first turns the name into an IP address.
2	Transport	TCP opens a connection (handshake) and puts destination port 80/443 and a random source port in the segment header.
3	Network	IP adds the source and destination IP address, making a packet; the router decides the next hop.
4	Data Link	Ethernet/Wi-Fi adds the MAC address of the next device (your router) and a CRC trailer, making a frame.
5	Physical	The frame is sent as bits (signals) on the cable or air.
6	Server side	The reverse: bits → frame → packet → segment → HTTP request; the web server replies in the same way.

5.6 OSI model vs TCP/IP model

● LEVEL 2 · EXAM

Point	OSI model	TCP/IP model
Developed by	ISO (1984)	DARPA (US Department of Defense)
Number of layers	7	4 (or 5)
Nature	Theoretical reference model; model first, protocols later	Practical; protocols first, model later
Use	Used for teaching and understanding	Used on the real Internet
Session and presentation	Separate layers	Merged into the Application layer
Services, interfaces, protocols	Clearly separated	Not clearly separated
Network layer	Connectionless and connection-oriented	Connectionless only (IP)
Transport layer	Connection-oriented only	Both: TCP (connection-oriented) and UDP (connectionless)
Flexibility	Strict layering	More flexible, layers can be skipped

QUICK TRICK

Remember the cross: **OSI network layer = both**, **OSI transport = connection-oriented only**; **TCP/IP network = connectionless only**, **transport = both**.

5.7 Common mistakes

● LEVEL 2 · EXAM

COMMON MISTAKE

- Mixing the order of layers. Use the mnemonics and always check whether the question asks top-down or bottom-up.
- Saying the router works at the data link layer. A router works at **Layer 3**; a switch at Layer 2.
- Saying TCP is at the network layer. TCP and UDP are at the **transport** layer; IP is at the network layer.
- Saying MAC addresses stay the same all the way. **MAC changes at every hop; IP stays the same.**
- Forgetting that the data unit names change: segment (4), packet (3), frame (2), bits (1).
- Saying the browser is the application layer. The browser is a program that **uses** an application-layer protocol (HTTP).
- Saying OSI is the model of the Internet. The Internet uses **TCP/IP**; OSI is the reference model.

Quick-Solve Tricks — Summary

Situation	Trick
OSI layers	7 layers, ISO, 1984. Bottom-up: Physical, Data link, Network, Transport, Session, Presentation, Application
Data units	Bits (1), Frame (2), Packet (3), Segment (4), Data (5-7)
Addresses	MAC (2), IP (3), Port (4), URL (7)
Devices	Hub / repeater: 1 · Switch / bridge: 2 · Router: 3
Routing, IP	Network layer
End-to-end, ports	Transport layer
Encryption, compression, translation	Presentation layer
Dialog control, checkpoints	Session layer
TCP/IP layers	Application (5-7), Transport (4), Internet (3), Network Access (1-2)
TCP vs UDP	TCP reliable, connection-oriented, 20-byte header · UDP fast, connectionless, 8-byte header
Handshake	SYN → SYN-ACK → ACK
MSS	$MTU - IP\ header - TCP\ header = 1500 - 20 - 20 = 1460$
Ports	16 bits: 0-1023 well-known; HTTP 80, HTTPS 443, FTP 21, SMTP 25, DNS 53
Across routers	IP stays the same; MAC changes every hop

BPSC QUICK CHECK — Chapter 5

OSI = 7 layers (ISO)	TCP/IP = 4 layers (DARPA)
Router → Network layer	Switch → Data link layer
TCP → Transport, reliable	UDP → Transport, unreliable
Port number → 16 bits	MAC → 48 bits · IPv4 → 32 bits
Presentation → encrypt, compress, translate	Session → dialog control, synchronisation

Traps: layer order · TCP/IP vs OSI layer count · MAC vs IP across routers · TCP vs UDP header size · segment, packet and frame names.

Practice questions

● LEVEL 1 · BEGINNER

- **Q1.** How many layers are there in the OSI model and in the TCP/IP model?
- **Q2.** Which layer converts a stream of bits into frames?
- **Q3.** Which layer is responsible for routing?
- **Q4.** Name the data units at the transport, network and data link layers.
- **Q5.** Which OSI layer handles encryption and compression?
- **Q6.** Which organisation developed the OSI model?

● LEVEL 2 • EXAM

- **Q7.** Write the seven OSI layers from the bottom to the top.
- **Q8.** Give three differences between TCP and UDP.
- **Q9.** Give the well-known port numbers of HTTP, HTTPS, SMTP, DNS and FTP (control).
- **Q10.** When a packet crosses several routers, which address changes at every hop: MAC or IP?
- **Q11.** Map the four TCP/IP layers to OSI layers.
- **Q12.** What is encapsulation?

● LEVEL 3 • BPSC / PGT

- **Q13.** Ethernet MTU = 1500 bytes, IP header = 20 bytes, TCP header = 20 bytes. Find the MSS and the largest Ethernet frame (header 14 + trailer 4 bytes).
- **Q14.** 5000 bytes of data are sent with TCP/IP over Ethernet using MSS = 1460. Find the number of segments, the total overhead and the total bytes sent (TCP 20 + IP 20 + Ethernet 18 per segment).
- **Q15.** A client starts a TCP connection with initial sequence number 100 and the server uses 300. Write the three handshake segments with seq and ack numbers.
- **Q16.** At which layer does each of these work: (a) CRC, (b) port number, (c) JPEG, (d) dialog control, (e) IP address, (f) voltage levels?
- **Q17.** The port field is 16 bits. How many ports exist and how many are well-known?
- **Q18.** A UDP datagram has a length field of 520. How many bytes of data does it carry?

Answers

Q	Solution
1	OSI: 7 layers. TCP/IP: 4 layers (some books show 5).
2	Data Link layer (framing).
3	Network layer (IP, routers).
4	Transport: segment (TCP) or datagram (UDP). Network: packet. Data link: frame.
5	Presentation layer.
6	ISO (International Organization for Standardization).
7	Physical, Data Link, Network, Transport, Session, Presentation, Application.
8	TCP: connection-oriented, reliable with ACKs, ordered, 20-byte header, slower. UDP: connectionless, unreliable, no ordering, 8-byte header, faster.
9	HTTP 80, HTTPS 443, SMTP 25, DNS 53, FTP control 21 (data 20).
10	The MAC address changes at every hop. The IP address stays the same (except with NAT).
11	Application = OSI 7, 6, 5. Transport = 4. Internet = 3. Network Access = 2 and 1.
12	Adding a header (and at layer 2 a trailer) to the data at each layer as it moves down the stack at the sender.
13	MSS = 1500 – 20 – 20 = 1460 bytes . Largest frame = 1460 + 20 + 20 + 14 + 4 = 1518 bytes .
14	$5000 \div 1460 = 3.42$, so 4 segments ($1460 \times 3 + 620$). Overhead = $4 \times 58 = 232$ bytes. Total = $5000 + 232 = 5232$ bytes.
15	1. Client → server: SYN, seq = 100. 2. Server → client: SYN-ACK, seq = 300, ack = 101. 3. Client → server: ACK, seq = 101, ack = 301.
16	(a) Data Link, (b) Transport, (c) Presentation, (d) Session, (e) Network, (f) Physical.
17	$2^{16} = 65\,536$ ports (0–65535). Well-known: 0–1023 = 1024 ports.
18	The UDP header is 8 bytes and the length covers header + data, so data = $520 - 8 = 512$ bytes.

REMEMBER

Next: Chapter 6 — Network Protocols (HTTP and HTTPS, FTP, TFTP, SMTP, POP3, IMAP, TELNET, SSH, PPP, VoIP, DNS, DHCP, SNMP, with how each works).

Roadmap: Ch 7 IP Addressing and Subnetting · Ch 8 Web Services (WWW, HTML, XML, URL, domain names, hosting) · Ch 9 Network Security · Ch 10 Mobile Telecom, Cloud and IoT.